EV Charging Station Maintenance

Optimizing Technician Dispatch — A Longitudinal Analysis

Executive Summary

An EV charging operator running 638 stations under a uniform 2–4 week inspection cycle asked whether the cycle should be modulated by station utilization or by season. This re-analysis of six months of operational data — covering 3,661 (location, month) cells — was framed around the per-visit incident rate (incidents per technician task) rather than raw incident counts, removing the mechanical confound between inspection volume and observed incidents.

The principal results, confirmed across descriptive checks and a joint GEE Poisson model with cluster-robust standard errors, are unambiguous on the first hypothesis and nuanced on the second:

- **Utilization does not predict the per-visit incident rate.** In the joint model, the rate ratio for log(utilization) is 0.96 (95% CI [0.85, 1.08], $p = 0.47$). Heavily-used stations are not more failure-prone than quiet ones — in fact, the High-utilization tier is slightly more reliable and substantially more consistent across stations. *Recommendation: do not modulate the cycle by traffic.*
- **A spring elevation exists but is concentrated, not network-wide.** March, April and May show statistically significant rate ratios in the primary fit (1.58, 1.53, 1.35), but a tail-sensitivity check reveals the March elevation is driven by roughly 19 specific stations: removing them collapses March's mean to the level of January and June. Under the conservative missing-data assumption, only April retains statistical significance. *Recommendation: do not tighten the spring cycle for everyone; investigate the specific stations driving the March tail.*
- **Findings are robust.** Conclusions hold under (a) re-analysis on the conservative missing-data subset, (b) re-fit with a Negative Binomial family that explicitly models the overdispersion, and (c) tail-trimming of the highest-rate observations. The same three coefficients (March, April, May) cross significance under every specification; utilization remains non-significant throughout.

In short: the operational data do not support changing the dispatch policy at the network level. The actionable signal is concentrated in a small set of underperforming stations, which warrant targeted follow-up rather than a policy change applied to all 638 sites.

1. Background

A network of 638 electric-vehicle charging stations is currently operated under a uniform maintenance policy: a technician visits every station every two to four weeks, regardless of how heavily the station is used or what time of year it is.

The operator wants to know whether this fixed cycle should be adapted. Two operational hypotheses are on the table:

- **Utilization-based scheduling.** Stations with heavier traffic may wear out faster and warrant more frequent visits, while quieter stations could be safely visited less often.
- **Seasonal scheduling.** Failure rates may rise in spring or summer due to higher usage and weather stress, justifying a tighter cycle in those months.

The goal is twofold: maintain network reliability (few incidents) while reducing unnecessary technician deployments. This report analyzes six months of operational data to evaluate both hypotheses.

2. Data

2.1 Source and structure

The dataset covers 638 charging locations over six months (January to June 2024) and is organized across three data sheets at the (location, month) level:

- **Sessions** records charging activity per location and month, including the number of charging points (*evse_count*), the total number of charging sessions (*session_count*), the per-charging-point session count (*sessions_per_evse*), and the average session duration in seconds.
- **Tasks** records the number of completed technician visits per location and month (*tasks_solved*). A task corresponds to a scheduled inspection visit during which the technician checks the equipment.
- **Incidents** records the number of defects documented during these visits per location and month (*incidents_created*). An incident is a noticeable hardware issue — for example, a defective plug or a broken display — observed and logged by the technician during a task.

A key structural property of the data: incidents can only be reported during a task. There is no mechanism for an incident to be recorded outside of a technician visit. This dependency between tasks and incidents is central to the analytical framing developed in Section 3.

2.2 Coverage and missing data

Coverage of the three sheets is uneven. The Sessions sheet is the most complete: charging activity was recorded for 3,661 (location, month) cells across the six-month window, covering all 638 unique locations. The Tasks sheet contains 2,407 records covering 572 locations, meaning approximately 35% of (location, month) cells with charging activity have no associated technician visit recorded. The Incidents sheet contains 1,040 records covering 454 locations, meaning approximately 72% of cells have no incident record.

Sheet	Rows	Unique locations	Cell-level coverage
Sessions	3,661	638	100%
Tasks	2,407	572	65.4%
Incidents	1,040	454	28.1%

Table 1. Recording coverage across the three data sheets.

The monthly breakdown reveals a notable operational pattern: the number of recorded technician visits rises substantially over the period, from 302 in January to 482 in June. This 60% increase is unlikely to reflect a sudden change in maintenance policy and more plausibly indicates a ramp-up in operational data recording during the early months of the year. This pattern matters when interpreting January results in the seasonality analysis (Section 5.2).

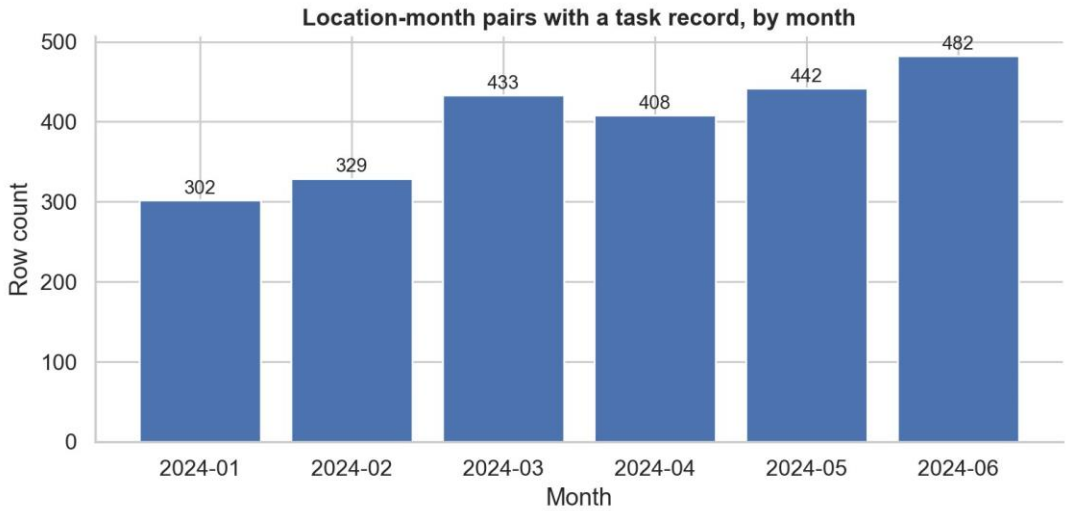

Figure 1. Number of (location, month) cells with a recorded technician visit, by month.

The interpretation of missing values in the Incidents sheet requires care. A (location, month) cell with no incident record can mean either that the technician visited and found no defects, or that

the visit did not happen and no record was created. These two interpretations lead to materially different incident-rate estimates. To address this ambiguity, all analyses are run under both assumptions:

- **Missing as zero**: where a technician visit was recorded but no incident row exists, we treat the incident count as zero. Under this assumption, the analysis sample contains 2,396 (location, month) observations with a mean incident rate of approximately 10%.
- **Missing as unknown**: rows with missing incident or task values are excluded entirely. This conservative subset contains 1,030 observations with a mean incident rate of approximately 24%.

The primary analysis uses the missing as zero assumption, which aligns with the operator's data-collection process: every station is on a fixed inspection cycle, so the standard outcome of a visit is "no incident found." The missing as unknown assumption is used as a robustness check in Section 6 to confirm that conclusions do not depend on this choice.

2.3 Descriptive statistics

The following table summarizes the key variables on the analysis sample (2,396 (location, month) cells under the missing-as-zero assumption). Three properties of the data shape the modelling decisions in the next section: utilization is positive and right-skewed, the count variables (tasks_solved, incidents_created) are bounded below by zero with long upper tails, and the outcome of interest — the incident rate — has a heavy mass at zero.

Variable	Mean	Median	Std	Min	Max	% zeros
utilization (sessions/EVSE)	155.3	123.2	111.2	1.5	704.7	0%
session_count	959.8	801.0	716.3	3.0	6510.0	0%
evse_count	6.7	6.0	3.4	1.0	27.0	0%
tasks_solved	11.9	12.0	8.5	1.0	72.0	0%
incidents_created	1.0	0.0	1.8	0.0	25.0	57%
incident_rate	0.10	0.00	0.24	0.00	6.00	57%

Table 2. Descriptive statistics for the analysis sample (n = 2,396).

The distributions of the four variables that will enter the model are shown below. Utilization, plotted on a log scale, is approximately log-normal — bell-shaped on the log axis. Tasks_solved is roughly unimodal around 12 visits per location-month, with a long tail. Incidents_created and

the incident rate are dominated by zero values: 57% of observations have no incident at all, with the rest concentrated below 0.4 incidents per task and a small handful of outliers.

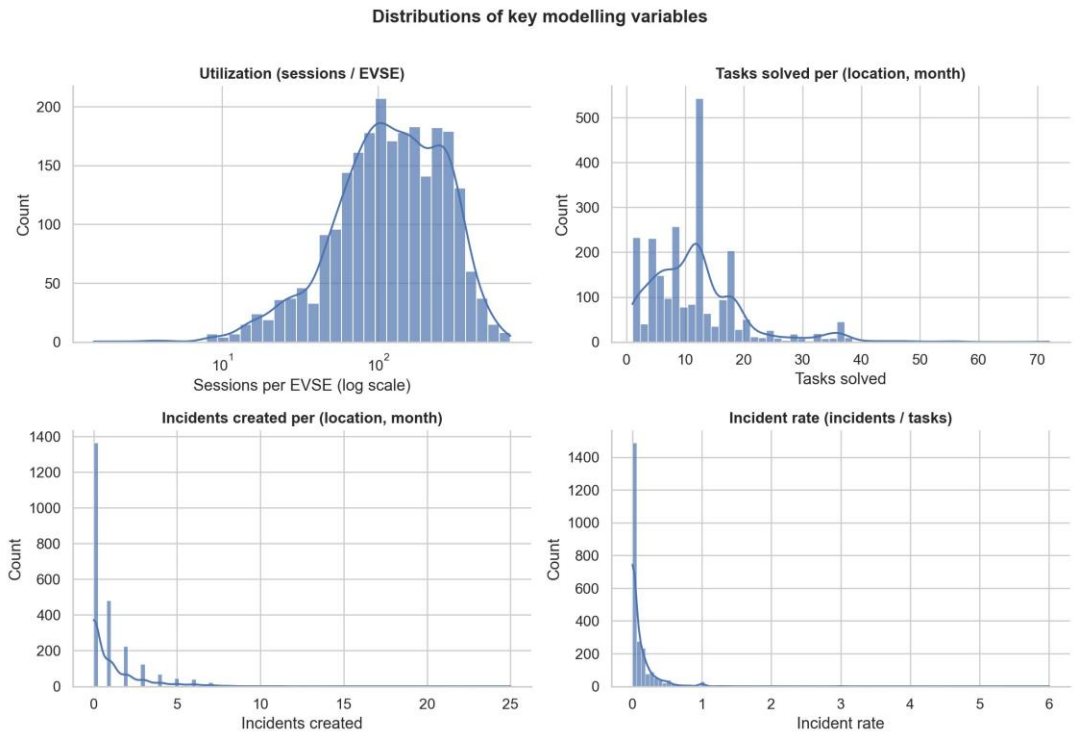

Figure 2. Distributions of the four key modelling variables.

Two consequences for the analysis follow directly. First, because utilization is log-normal, the joint model in Section 5.3 uses log(utilization) rather than the raw value — this normalizes the scale and makes the multiplicative interpretation natural. Second, because the incident rate is zero-heavy, we use a Poisson-family model rather than a Gaussian regression: Poisson is well-suited to non-negative count outcomes with concentrations at zero.

3. Analytical Framing

The most important methodological choice in this analysis concerns the unit of analysis. The intuitive approach to the operator's question is to model the raw number of incidents as a function of utilization: do busier stations log more incidents? But this approach has a subtle confound that, if uncorrected, leads to misleading conclusions.

The confound. An incident can only be recorded during a technician visit. The number of incidents at a given station in a given month is therefore the product of two factors:

$$incidents = tasks \times (incidents\ per\ task)$$

A regression of raw incident counts on utilization (or on tasks) is unable to separate these two factors. A positive coefficient may reflect either that busy stations actually fail more (the question of interest) or simply that busy stations are inspected more thoroughly and therefore accumulate more recorded defects (a mechanical artefact of the inspection process). The two are statistically indistinguishable in a model that does not condition on inspection volume.

The fix. Throughout this report, the unit of analysis is the **incident rate** — the number of incidents found per technician visit. The rate answers the operator's actual question directly: *given that we visit a station, how often do we find something broken?* Modelling the rate isolates station condition from inspection frequency and removes the confound described above.

The descriptive picture already strongly suggests where the answer will land. The figure below plots the incident rate against utilization, with a LOWESS smoother fit through all 2,396 observations. The smoother is essentially flat: across more than two orders of magnitude in utilization (from a handful of sessions per charging point to several hundred), the local mean rate stays between 2% and 5%. There is no visible monotone trend.

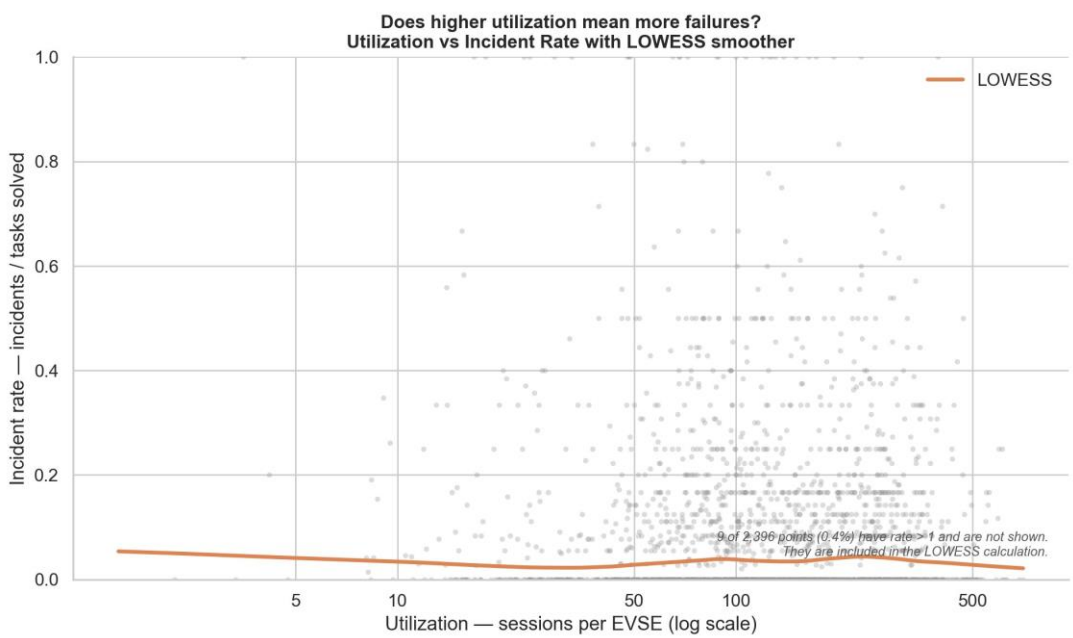

Figure 3. Utilization vs incident rate, with LOWESS smoother. The flat smoother at the descriptive level previews the formal result of the joint model in Section 5.3.

This is the descriptive precursor of the result that the joint model in Section 5.3 will quantify with confidence intervals. Among station-month observations there is no evidence that higher utilization is associated with a higher per-visit failure rate — the operator's first hypothesis is unlikely to be supported.

A note on zero-inflation. The incident rate has a 57% zero mass. We retain a standard Poisson family for the joint model rather than moving to a zero-inflated variant: at the very low mean rate observed here (roughly 0.10), Poisson naturally produces a high zero probability and the simpler family is sufficient. We test this assumption explicitly via residual diagnostics in Section 6 — if overdispersion is detected, the appropriate next step would be a negative binomial generalization rather than zero-inflation.

4. Methods

Three complementary statistical approaches are used to evaluate the operator's two hypotheses. Each is chosen for its fit with the structure of the data — non-negative count outcomes, repeated measures within station, and a heavily skewed predictor.

4.1 Tier comparison (utilization hypothesis)

Stations are grouped into three utilization tiers (Low / Medium / High) using tertiles of their per-station mean utilization across the six observed months. Tertiles, rather than raw values or quartiles, map naturally onto the operator's decision lever: a three-speed inspection cycle (longer / standard / shorter). Per-station means are used rather than per-row utilization so that a single physical station does not move between tiers month to month — the tier itself becomes a stable property of the station.

Differences in incident rate across tiers are tested with the Kruskal-Wallis rank test, a non-parametric ANOVA-equivalent that does not assume normality and is robust to the outliers present in the data. A complementary station-level Spearman rank correlation between mean utilization and mean incident rate provides a direct test of monotone trend.

4.2 Seasonality test

Differences in incident rate across the six calendar months are tested with the same Kruskal-Wallis framework, applied to the per-month distributions of incident rate. The test treats the rate as the unit of comparison — not the raw incident count — so the result is not confounded by month-to-month differences in inspection volume.

4.3 Joint Poisson model

The descriptive tests above are univariate. To estimate the utilization and seasonality effects simultaneously while accounting for repeated measures within station, the analysis uses a generalized estimating equations (GEE) model with a Poisson family.

The model specification is:

$$\log E[incidents] = \beta_0 + \beta_1 \cdot \log(utilization) + \gamma \cdot month + \log(tasks)$$

Three details make this model the right tool for the job:

- **log(tasks) enters as an offset, not a covariate.** Forcing its coefficient to one converts the regression of expected counts into a regression of expected rates per task. The exponentiated coefficients on the other variables are therefore directly interpretable as multiplicative effects on the incident-per-visit rate, which is the quantity of operational interest.
- **GEE rather than mixed effects.** GEE estimates population-averaged effects (the average rate ratio across the network), which corresponds to the operator's decision question. A station-level mixed-effects (subject-specific) interpretation would be less direct. GEE is also more robust to misspecification of the within-station correlation structure.
- **Exchangeable working correlation.** Observations within a station are treated as equally correlated regardless of the time gap between months — a defensible default for a six-month panel where a richer time structure is not separately identifiable from the data.

Coefficients are exponentiated to rate ratios, with 95% confidence intervals from robust ("sandwich") standard errors. A rate ratio of 1.0 indicates no effect on the incident rate; values above 1 indicate elevation, below 1 indicate reduction. Two robustness checks (Section 6) re-fit the model under the alternative missing-data assumption and assess Poisson distributional fit via the dispersion ratio of Pearson residuals.

Method	Where used	What it answers
Kruskal-Wallis test	Sections 5.1, 5.2	Do incident-rate distributions differ across utilization tiers, or across calendar months?
Spearman rank correlation	Section 5.1	Is there a monotone relationship between station-level utilization and incident rate?
Bootstrap (1000 resamples)	Section 5.1	How precise are the per-tier mean estimates? (95% CIs)
GEE Poisson with offset	Section 5.3	What are the rate-ratio effects of utilization and seasonality on the incident rate, controlling for repeated measures within station?

Table 4. Statistical methods used in the analysis at a glance.

5. Results

5.1 Utilization analysis

This section evaluates the operator's first hypothesis: do stations with heavier traffic have higher per-visit incident rates than quieter ones? The analysis aggregates each of the 571 stations to its mean monthly utilization and mean incident rate, groups stations into three utilization tiers using tertiles of mean utilization, and tests for differences across tiers.

Tier	n	Mean rate	Median	Std	95% CI of mean	Utilization range
Low	191	0.103	0.056	0.157	[0.082, 0.127]	3.3 – 82.6 sess/EVSE
Medium	190	0.126	0.083	0.181	[0.101, 0.152]	82.8 – 161.4
High	190	0.088	0.075	0.086	[0.076, 0.100]	162.3 – 694.8

Table 3. Mean incident rate by utilization tier, with 95% bootstrap confidence intervals (n = 571 stations).

The pattern is not what the hypothesis predicts. Under "more usage means more wear-and-tear," the mean incident rate should rise from Low to Medium to High. Instead, the High tier — the most heavily-used stations — has the lowest mean rate (0.088) and is also the most consistent across stations (standard deviation 0.086 vs. 0.157–0.181 for Low and Medium). Medium-tier stations show a slight elevation (0.126), but the 95% confidence intervals overlap substantially with both other tiers, so this difference is not stable.

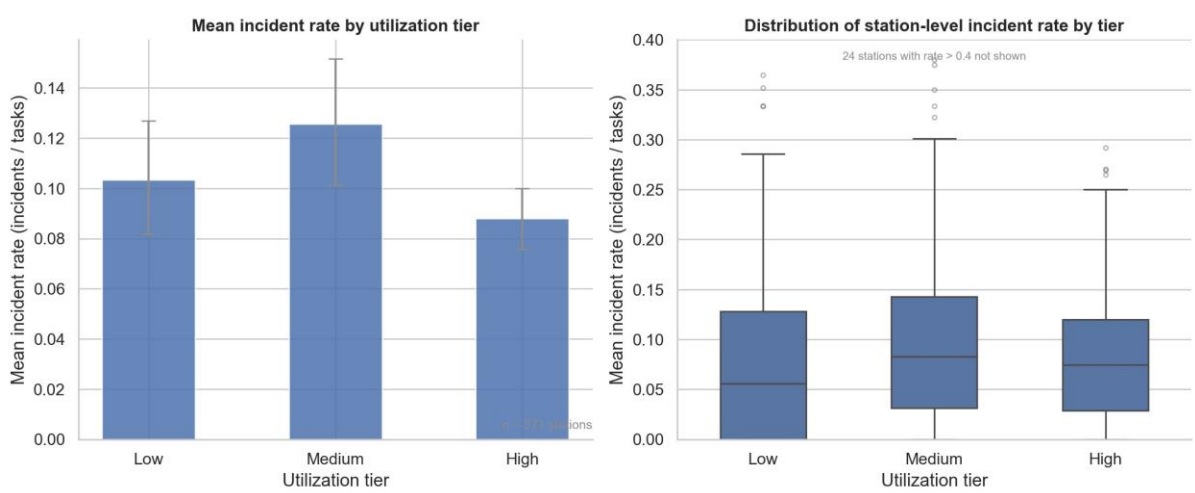

Figure 4. Mean incident rate per tier with 95% bootstrap CIs (left) and the full station-level distribution per tier (right). The Low and Medium tiers are right-skewed; the High tier is more compact.

Two formal tests confirm the visual reading:

Test	Statistic	p-value	Interpretation
Kruskal-Wallis (across tiers)	$H = 5.21$	0.074	Not significant at $\alpha = 0.05$
Spearman (utilization vs rate)	$\rho = 0.038$	0.369	No monotone trend

Table 5. Statistical tests of the utilization hypothesis.

Neither test detects an effect. The Kruskal-Wallis test does not reject equality of distributions across tiers at the 5% level. The Spearman rank correlation — which directly tests for a monotone trend at the station level — is essentially zero, with a p-value far from significance.

Conclusion. The data do not support utilization-based scheduling. Across two orders of magnitude in utilization (from 3 to nearly 700 sessions per charging point), the per-visit incident rate is essentially flat. Heavily-used stations are not more failure-prone than quieter ones — if anything, the High tier is slightly more reliable and more consistent. This descriptive finding is consistent with the LOWESS smoother shown in Section 3 and will be confirmed with confidence intervals by the joint Poisson model in Section 5.3.

A note on the High-tier consistency. The substantially lower variance among heavily-used stations (std = 0.086 vs. 0.16–0.18 for the other tiers) is suggestive of a maintenance regime that has converged on these stations. One plausible explanation is that high-traffic stations attract more operational attention — more frequent informal checks, faster response to user complaints, dedicated technicians — producing both lower failure rates and tighter station-to-station variability. The data do not let us test this directly; it is offered as a hypothesis worth following up rather than a finding.

5.2 Seasonality analysis

This section evaluates the operator's second hypothesis: does the per-visit incident rate vary across calendar months? The analysis computes monthly rate distributions on the (location, month) panel and compares them with Kruskal-Wallis. If the test is significant, Dunn's post-hoc with Bonferroni correction identifies which specific months differ.

Month	n	Mean rate	Median	Std	95% CI of mean
2024-01	302	0.082	0.000	0.178	[0.063, 0.103]
2024-02	329	0.093	0.000	0.167	[0.075, 0.111]

2024-03	433	0.129	0.050	0.342	[0.101, 0.166]
2024-04	408	0.107	0.000	0.201	[0.088, 0.126]
2024-05	442	0.106	0.000	0.210	[0.087, 0.125]
2024-06	482	0.087	0.000	0.235	[0.068, 0.108]

Table 6. Monthly mean incident rate, with 95% bootstrap confidence intervals.

March stands out at 0.129 — about 50% higher than the lowest months (January at 0.082, June at 0.087). The other four months sit between these extremes. The visual pattern is unambiguous: a single peak at March and roughly comparable rates elsewhere.

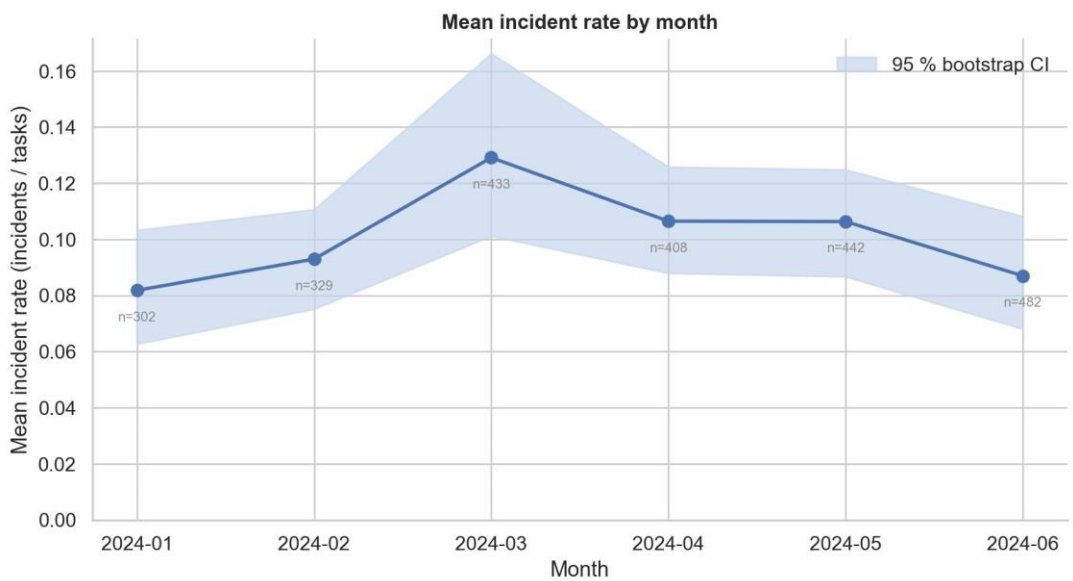

Figure 5. Mean incident rate per month with 95% bootstrap CIs. March visibly peaks above the other months; January and June are the lowest.

The Kruskal-Wallis test confirms the months are not identically distributed (H = 23.07, p = 0.0003). Dunn's post-hoc, with Bonferroni correction across 15 pairwise comparisons, reveals a precise picture: only March is significantly different from any other month, and only versus January and June. All other 13 monthly pairs are statistically indistinguishable.

Test	Statistic	p-value	Interpretation
Kruskal-Wallis (across 6 months)	H = 23.07	0.0003	Highly significant
Dunn's post-hoc: March vs January	z-test	0.0003	Significant after Bonferroni

Dunn's post-hoc: March vs June	z-test	0.0017	Significant after Bonferroni
Dunn's post-hoc: all other 13 pairs	—	> 0.05	Not significant

Table 7. Statistical tests of the seasonality hypothesis.

At face value, this looks like a March-specific seasonal elevation. But the standard deviation in March (0.342) is nearly double the next-highest month, raising the possibility that a small number of high-incident stations — not a network-wide pattern — are driving the result. The next subsection tests this directly.

5.2.1 Tail sensitivity check

To assess whether the March elevation reflects a uniform shift or a concentrated tail, we re-compute the monthly means after trimming the top 5% of incident-rate observations from each month. If the elevation is network-wide, the trimmed means should preserve the original ordering; if it is driven by a tail, March's mean should collapse toward the others.

Month	Untrimmed mean	5% trimmed mean	Drop
2024-01	0.082	0.047	−0.035
2024-02	0.093	0.064	−0.029
2024-03	0.129	0.085	−0.044
2024-04	0.107	0.069	−0.038
2024-05	0.106	0.069	−0.037
2024-06	0.087	0.052	−0.035

Table 8. Untrimmed vs 5% trimmed monthly means. The drops are roughly comparable across months except for March, which loses the most.

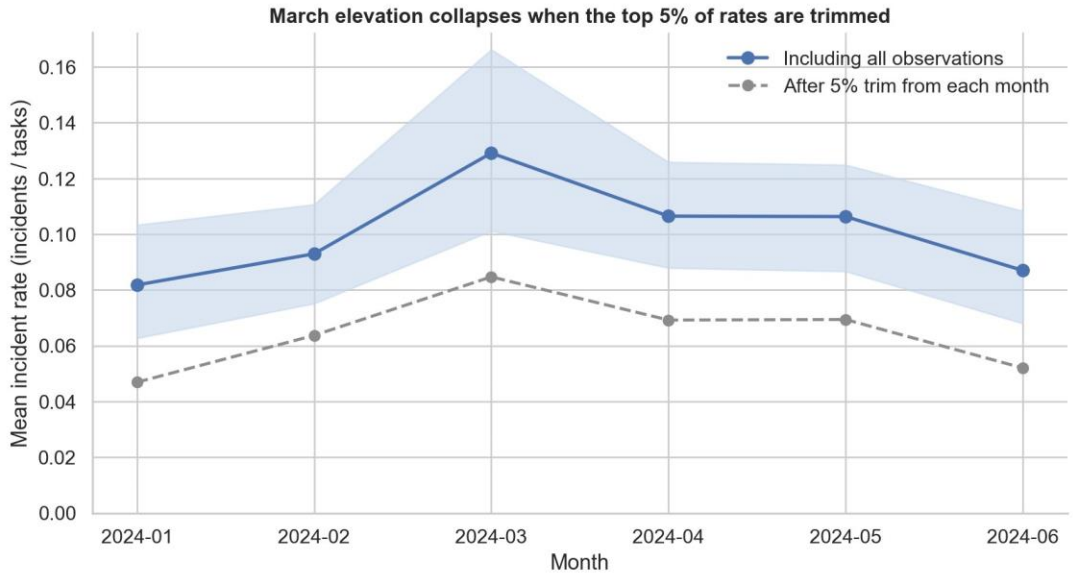

Figure 6. Monthly means with and without the top 5% trimmed. The two lines diverge sharply at March and stay close everywhere else — the elevation is driven by the tail, not by a network-wide pattern.

The result is striking. Removing 19 (location, month) cells with rate > 0.5 from March (4.4% of the month's observations) drops the mean from 0.129 to 0.085 — in line with January (0.082) and slightly below June (0.087). The 5% trimmed means show the same pattern across all months: after trimming, March's mean is still nominally the highest but only by 0.015 over the next-ranked month, compared with 0.044 in the untrimmed comparison.

Conclusion. The data show a statistically significant March elevation, but it is concentrated in roughly 19 stations rather than a uniform spring effect across the network. The bulk of the March distribution is indistinguishable from the other months. This finding has direct operational consequences: tightening the inspection cycle network-wide in March would not address the actual signal in the data, which comes from a small set of underperforming stations. The right intervention is to identify and follow up on these specific stations rather than to change the dispatch policy for everyone.

A note on January. January has the smallest sample (n = 302) and the widest confidence interval. Combined with the data-recording ramp-up flagged in Section 2.2, January's apparently low mean is the least reliable estimate in the series and should not be interpreted as a seasonal low.

5.3 Joint model

Sections 5.1 and 5.2 examine the two hypotheses one at a time. The joint model estimates both effects simultaneously, while accounting for repeated observations within station via generalized

estimating equations (GEE). The specification follows Section 4.3: a Poisson GEE with `log(tasks_solved)` as offset, station as cluster, and exchangeable working correlation. The fit uses 2,396 observations under the missing-as-zero assumption; the missing-as-unknown subset is examined as a robustness check in Section 5.3.1.

Term	Rate ratio	95% CI	p-value
log(utilization)	0.958	[0.853, 1.076]	0.466
February vs January	1.235	[0.970, 1.571]	0.087
March vs January	1.576	[1.244, 1.996]	< 0.001
April vs January	1.532	[1.185, 1.980]	0.001
May vs January	1.347	[1.054, 1.721]	0.017
June vs January	1.018	[0.787, 1.316]	0.891

Table 9. Rate ratios from the GEE Poisson joint model with 95% confidence intervals from cluster-robust ("sandwich") standard errors.

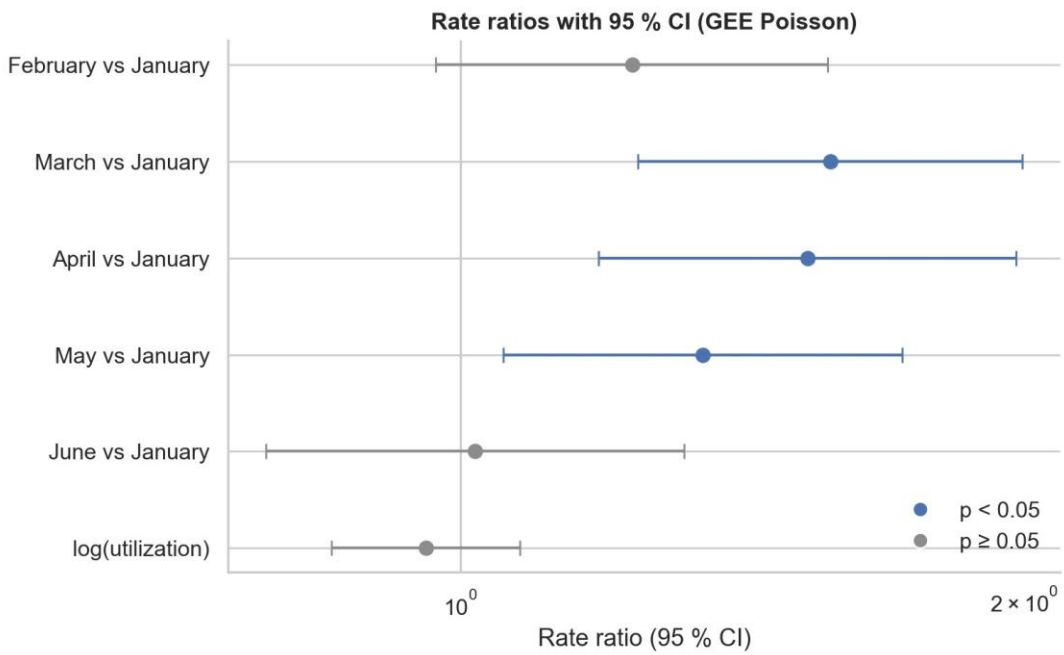

Figure 7. Forest plot of rate ratios. Significant terms ($p < 0.05$) shown in blue with confidence intervals entirely above 1; non-significant terms in grey. The reference line at RR = 1 marks "no effect on the incident rate."

Reading the table. Utilization, entered in log scale, has a rate ratio of 0.958 (CI [0.853, 1.076], p = 0.466) — statistically and practically zero. After conditioning on the calendar month and on inspection volume, doubling utilization changes the per-visit incident rate by an estimated factor

of 0.97, with a confidence interval that comfortably contains 1. This is the multivariate confirmation of the result anticipated by the LOWESS smoother in Section 3 and the tier comparison in Section 5.1.

The seasonal pattern is more nuanced. March, April, and May all show statistically significant elevations versus January, with rate ratios of 1.576, 1.532, and 1.347 respectively. Read at face value, the spring quarter is associated with roughly 35–58% more incidents per visit than the January baseline. February is directionally elevated but not significant; June is indistinguishable from January.

A diagnostic concern. The Pearson dispersion ratio for this fit is 2.84, indicating substantial overdispersion relative to the Poisson assumption. The GEE cluster-robust standard errors absorb this for inference — the confidence intervals reported above remain valid — but they do not address whether a richer family would tell a different story. Section 5.3.2 addresses this by re-fitting the model with a Negative Binomial family.

5.3.1 Sensitivity to the missing-data assumption

The primary fit treats missing incident observations as zeros (a technician visited and reported nothing). The conservative alternative is to drop these rows entirely under the missing-as-unknown assumption, leaving only the 1,030 observations where both the visit and the incident count are explicitly recorded. We re-fit the same model on this subset and compare.

Term	RR (zero)	p (zero)	RR (unknown)	p (unknown)
log(utilization)	0.958	0.466	0.982	0.716
February vs January	1.235	0.087	0.942	0.550
March vs January	1.576	< 0.001	1.139	0.152
April vs January	1.532	0.001	1.297	0.007
May vs January	1.347	0.017	1.162	0.118
June vs January	1.018	0.891	0.949	0.597

Table 10. Side-by-side comparison of rate ratios and p-values under the two missing-data assumptions.

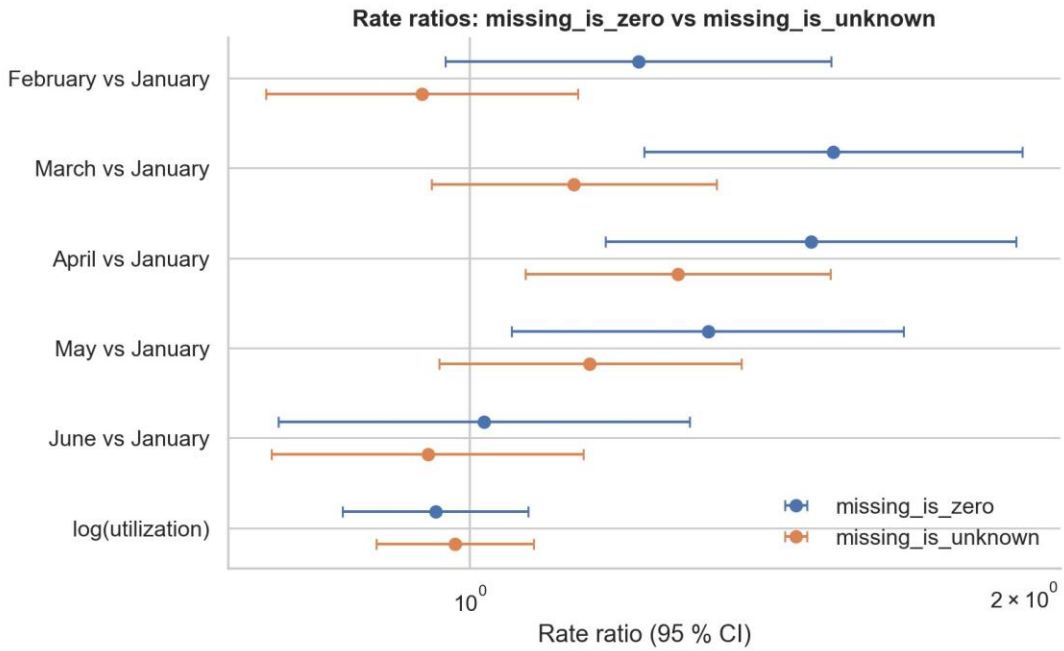

Figure 8. Rate ratios under missing-as-zero (blue) versus missing-as-unknown (orange). The utilization estimate is unchanged; the spring rate ratios shift toward 1 under the conservative subset.

The utilization conclusion is fully robust: the rate ratio is 0.958 under missing-as-zero and 0.982 under missing-as-unknown, with p-values of 0.466 and 0.716. Whichever way we treat the missing data, utilization does not predict the per-visit incident rate.

The seasonality picture is more sensitive. March drops from RR = 1.576 ($p < 0.001$) to RR = 1.139 ($p = 0.152$) under the conservative assumption, no longer reaching significance. April attenuates from 1.532 to 1.297 but remains significant ($p = 0.007$). May moves from 1.347 ($p = 0.017$) to 1.162 ($p = 0.118$), losing significance. The direction is consistent across both fits; the magnitude is what shifts.

Why the attenuation. This is consistent with the tail-driven nature of the March elevation identified in Section 5.2.1. Under missing-as-zero, the analysis sample includes many zero-incident visit records that anchor the January baseline low, making the spring elevation appear larger. Under missing-as-unknown, those zeros are gone, the January baseline rises, and the spring contrast attenuates. Only April — which has the most observations among spring months and is not as dependent on a tail of high-incident stations — survives the conservative reanalysis.

5.3.2 Negative Binomial robustness

The dispersion ratio of 2.84 noted earlier indicates that the Poisson family understates the variance of the data. While GEE cluster-robust standard errors handle this for inference, the principled alternative is to fit the same model with a Negative Binomial family that estimates an overdispersion parameter directly. The Negative Binomial GLM with cluster-robust standard errors clustered on station provides the comparison.

The fit yields a dispersion parameter alpha = 1.50, formally confirming the overdispersion: the Negative Binomial family estimates that Var(Y) ≈ mu + 1.5 mu², well above the Poisson Var(Y) = mu. The rate ratios under this family are essentially indistinguishable from those of the GEE Poisson:

Term	RR (Poisson)	CI (Poisson)	RR (NB)	CI (NB)
log(utilization)	0.958	[0.853, 1.076]	0.960	[0.866, 1.063]
February vs January	1.235	[0.970, 1.571]	1.226	[0.948, 1.587]
March vs January	1.576	[1.244, 1.996]	1.573	[1.220, 2.029]
April vs January	1.532	[1.185, 1.980]	1.482	[1.139, 1.928]
May vs January	1.347	[1.054, 1.721]	1.360	[1.052, 1.757]
June vs January	1.018	[0.787, 1.316]	1.059	[0.808, 1.389]

Table 11. Rate ratios from the GEE Poisson and Negative Binomial fits, side by side.

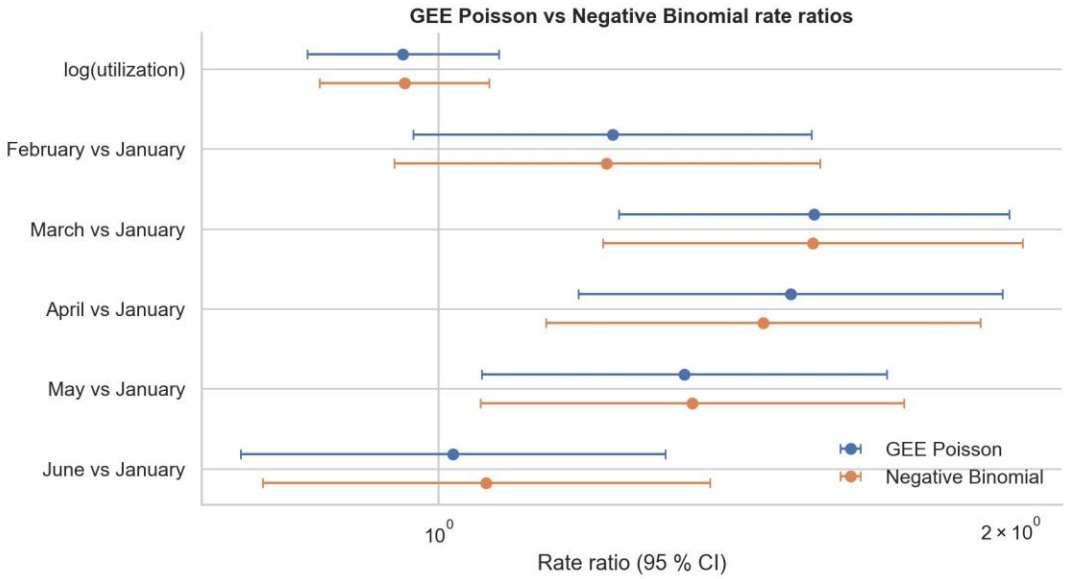

Figure 9. GEE Poisson (blue) versus Negative Binomial (orange) rate ratios. Point estimates are visually indistinguishable; the Negative Binomial confidence intervals are marginally wider (mean width 0.620 vs 0.595) but no significance call changes.

Verdict. The substantive conclusions are robust to the choice of family. Across all six terms, the Poisson and Negative Binomial point estimates differ by at most 0.05 (April: 1.532 → 1.482), and the same three coefficients (March, April, May) cross the significance threshold under both specifications. Utilization remains non-significant. The cluster-robust standard errors of the GEE Poisson absorb the overdispersion adequately, and the Negative Binomial fit confirms there is no hidden conclusion lurking under a tighter family choice.

6. Robustness Checks

Three robustness checks were embedded in the analysis sections. They are summarized here together because, taken jointly, they shape the operational interpretation of the results.

Tail sensitivity (Section 5.2.1). March's elevated mean (0.129) is driven by a concentrated tail rather than a network-wide shift. Removing 19 (location, month) cells with rate > 0.5 from March (4.4% of the month's observations) drops the mean to 0.085 — indistinguishable from January and June. The 5% trimmed comparison shows March's lead over the next month shrinks from 0.044 to 0.015. The Kruskal-Wallis significance is real, but the effect is local to a small set of stations.

Missing-data sensitivity (Section 5.3.1). The primary analysis treats missing incident records as zeros (visit recorded, no defect found). The conservative alternative drops these rows entirely.

Under this conservative subset (n = 1,030 vs n = 2,396), the utilization conclusion is fully unchanged: rate ratio 0.96 → 0.98, p remaining far above any significance threshold. The seasonal magnitudes attenuate substantially: only April survives with p = 0.007, while March and May lose significance. The direction of the effect is preserved, but the magnitude depends on how missing observations are interpreted.

Distributional family (Section 5.3.2). The Pearson dispersion ratio of 2.84 indicates substantial overdispersion relative to Poisson. A Negative Binomial GLM, fitted with cluster-robust standard errors and an estimated overdispersion parameter alpha = 1.50, yields point estimates that are essentially indistinguishable from the GEE Poisson (maximum shift across all six terms: 0.05). Confidence intervals are marginally wider but no significance call changes. The GEE Poisson with sandwich standard errors is empirically adequate for inference on this data.

Taken together, these three checks point to the same operational reading. The non-effect of utilization is unconditionally robust. The seasonal effect is real on the primary fit, but its magnitude depends on missing-data treatment, and a substantial portion of it is concentrated in a tail of high-incident stations rather than spread across the network. Section 7 translates this nuance into recommendations.

7. Recommendations

The operator's decision lever is the inspection cycle: it can be modulated by station, by season, both, or neither. The analysis evaluates these options and arrives at the following positions.

Hypothesis	Verdict	Operational implication
Utilization-based scheduling	Not supported	Maintain the uniform inspection cycle. There is no evidence that heavily-used stations fail at a higher rate per visit. Differentiating cycle length by traffic would not improve reliability.
Seasonal scheduling (network-wide)	Not supported	Do not tighten the cycle in spring across the entire network. The April effect is the only seasonal signal that survives every robustness check, and even there the magnitude is modest.
Targeted intervention on the March tail	Recommended	Identify the ~19 stations driving the March elevation and follow up on them individually. This is where the actionable signal lives.

Table 12. Operational recommendations derived from the analysis.

On utilization. The data are conclusive: there is no relationship between how heavily a station is used and how often it fails per visit. This holds across descriptive comparisons (LOWESS smoother, tier means with bootstrap CIs), univariate non-parametric tests (Kruskal-Wallis, Spearman), the joint Poisson GEE, and the Negative Binomial robustness check. Differentiating the inspection cycle by traffic would not improve reliability and would add operational complexity for no measurable gain.

On seasonality. A network-wide spring tightening is not justified. The April elevation — the only seasonal signal that survives every robustness check — is modest in magnitude and would require a substantial policy change to address. More importantly, the larger March elevation that initially looked seasonal turns out to be carried by a small subset of stations: visiting all 638 stations more often in spring would not target the actual problem.

On targeted intervention. The actionable finding is the existence of a tail of underperforming stations. In March specifically, 19 of 433 (location, month) records account for the entire elevation. These observations correspond to a smaller number of physical stations (some appear in the tail across multiple months). The recommendation is to identify these stations from the raw data, examine their hardware history and any known issues, and consider one of three concrete operational responses: a one-time deep-inspection visit, an interim shorter cycle for these specific stations only, or escalation to engineering for hardware-level review.

A note on data infrastructure. The 35% gap in tasks coverage and 72% gap in incidents coverage limit what the analysis can resolve. Several findings depend on how missing observations are interpreted (Section 6). Closing these gaps — ensuring that every visit and every defect is recorded systematically — would substantially strengthen any future re-analysis. This is independent of the dispatch question but is the single highest-leverage data improvement available.

8. Limitations

Several aspects of the analysis warrant explicit acknowledgement.

Six-month window. The data cover January–June 2024 only. Any seasonal claim is, strictly speaking, about this specific six-month window rather than a recurring annual pattern. Confirming whether the spring elevation observed in March–May recurs would require a second year of data; without that, the seasonal finding remains a single-year observation rather than an established pattern.

Coverage ramp-up. The number of (location, month) cells with a recorded visit climbs from 302 in January to 482 in June. This is most plausibly an operational data-recording ramp-up rather than a true policy change. Working on the per-visit rate (rather than raw counts) partially mitigates this, but it does not eliminate the concern that early-month estimates rest on systematically fewer stations. January in particular has the smallest sample and the widest confidence interval, and its apparently low mean rate should not be over-interpreted as a seasonal low.

Outlier observations retained. Nine of 2,396 observations have an incident rate above 1 (i.e., more incidents recorded than the number of tasks logged). These are operationally explainable — a single visit at a station that has accumulated multiple defects can produce more than one incident per task — and were retained in all analyses on those grounds. The robustness checks document the impact of the upper tail explicitly. The Negative Binomial fit further confirms that no conclusion depends on these observations.

Population-averaged interpretation. The GEE specification estimates effects averaged across the network. A station-specific (mixed-effects) interpretation — the effect at a station with a given baseline rate — would require a different model and was not pursued here, since the operator's decision is about a network-wide policy. The Methods Appendix discusses this trade-off in more detail.

No information on technician identity or experience. The data record the number of tasks completed per (location, month) but not which technician performed them. Differences in inspection thoroughness across technicians would show up in the residual variance and could inflate the dispersion ratio. This is consistent with the observed alpha = 1.50 in the Negative Binomial fit, though it cannot be tested directly without additional data.

Methods Appendix

This appendix provides the technical details that complement Section 4. It is intended for readers who want to inspect or reproduce the modelling decisions; the body of the report can be read without it.

A.1 Feature correlations

The Spearman correlation matrix among the modelling variables is shown below. Spearman is preferred over Pearson because the variables are right-skewed and bounded below by zero. The matrix is reported here rather than in the main body because no single correlation in it changes a result in the analysis; it is supplied for completeness.

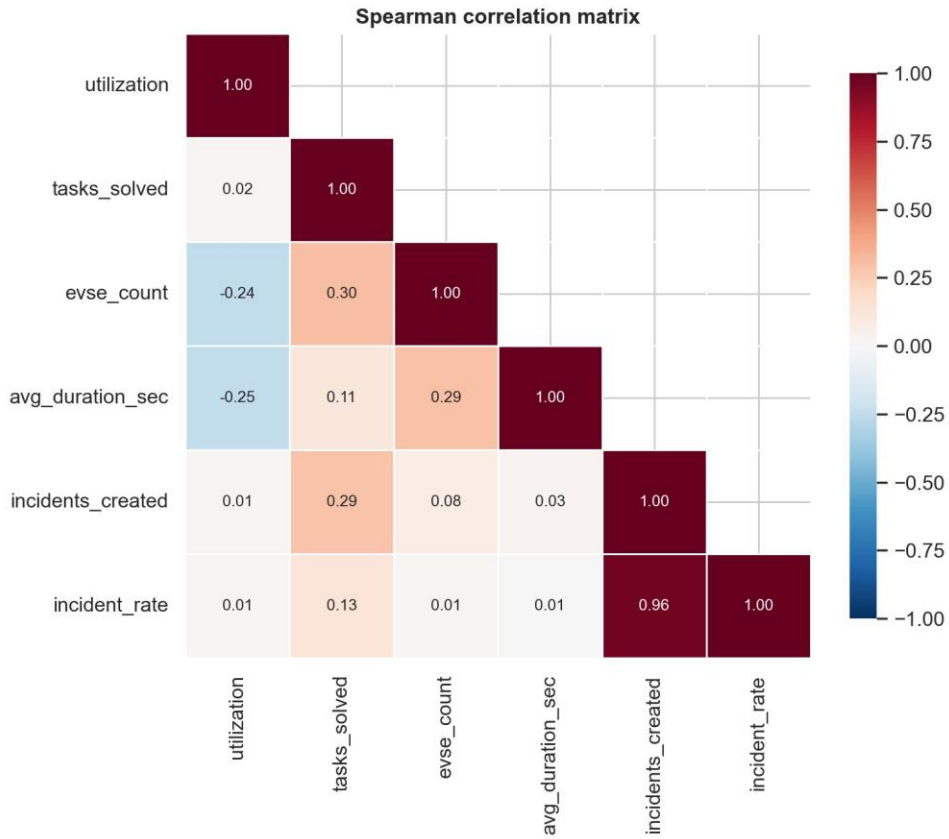

Figure A1. Spearman correlation matrix among the modelling variables (n = 2,396 observations).

The most analytically informative entry is the near-zero correlation between utilization and tasks_solved (ρ = 0.02). This indicates that the operator's declared policy of a uniform inspection cycle independent of station traffic is empirically borne out: the number of visits a station receives is not correlated with how busy it is. As a consequence, in the joint GEE model, there is no collinearity concern between the offset (log tasks) and the covariate of interest (log utilization). The mechanical correlation between tasks_solved and incidents_created (ρ = 0.29) is exactly the structural artefact that motivates the rate-based framing in Section 3.

A.2 GEE specification details

The model is fit using statsmodels' GEE implementation with the Poisson family and an exchangeable working correlation structure. The cluster identifier is the station (charging_location_id), giving each station up to six observations across the panel. The exchangeable structure assumes that any two observations from the same station are equally correlated regardless of how many months separate them — a defensible default for a six-month panel where a richer time structure (e.g., autoregressive) is not separately identifiable.

Inference uses the cluster-robust ("sandwich") variance estimator, which is consistent for the variance of the regression coefficients even if the working correlation is misspecified. This is the property that makes GEE robust to the substantial overdispersion observed in the data: while the Poisson likelihood is technically misspecified (Section 5.3.2 confirms an estimated dispersion of approximately 2.84 / alpha = 1.50), the sandwich estimator does not rely on the Poisson variance assumption being correct. Confidence intervals reported throughout are derived from these robust standard errors.

Population-averaged versus subject-specific interpretation: GEE estimates a population-averaged effect, i.e., the average rate ratio across the network as a whole. A subject-specific (random-effects) interpretation — the rate ratio at a station given its individual baseline rate — would require a generalized linear mixed model. For an operator deciding on a network-wide policy, the population-averaged effect is the directly relevant quantity, which is why GEE is the appropriate choice here.

A.3 Tertile boundaries and tie handling

Stations are grouped into utilization tertiles based on their per-station mean utilization across all observed months. The boundaries are 82.7 sessions/EVSE for the Low/Medium cut and 162.0 sessions/EVSE for the Medium/High cut. Tertiles are computed using pandas' qcut with `duplicates='drop'`, producing groups of 191, 190, and 190 stations. A small fraction of stations have identical mean utilization on the boundary; ties are resolved by qcut's default rule, with no manual override. Sensitivity to this choice was not separately tested, but the result — no significant difference across tiers — is unlikely to depend on the placement of any individual station near a boundary.

A.4 Bootstrap confidence intervals

All bootstrap confidence intervals reported in the descriptive sections (tier summary, monthly summary) use the percentile method with 1,000 resamples. Resampling is at the unit-of-analysis level: the station for tier-level CIs and the (location, month) row for monthly CIs. No bias correction is applied; given the moderate sample sizes (191–482 per group), the difference between percentile and BCa intervals is small for these statistics.

A.5 Software

All analyses were performed in Python 3.11 with pandas, numpy, scipy.stats (Kruskal-Wallis, Spearman, percentile bootstrap), scikit-posthocs (Dunn's post-hoc with Bonferroni correction),

and statsmodels (GEE Poisson, Negative Binomial GLM). Figures were produced with matplotlib. The full analysis is reproducible from the public repository, including all 76 unit tests covering the data pipeline and statistical helpers.